

DEEP LEARNING

Labwork 2 - Linear Regression

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1 Algorithm Implementation

In this labwork, we implemented linear regression from scratch using gradient descent optimization. The goal is to learn parameters w_0 and w_1 such that the model:

$$\hat{y} = w_1 x + w_0$$

best fits the given dataset.

1.1 Model Definition

The linear regression model is defined as:

$$f(x) = w_1 x + w_0$$

1.2 Loss Function

We use the Mean Squared Error (MSE) with a factor $\frac{1}{2}$ for convenience in derivative computation:

$$J(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n \frac{1}{2} (\hat{y}_i - y_i)^2$$

1.3 Gradient Descent

The parameters are updated using gradient descent:

$$w_0 := w_0 - r \frac{\partial J}{\partial w_0}$$

$$w_1 := w_1 - r \frac{\partial J}{\partial w_1}$$

where r is the learning rate.

The partial derivatives are:

$$\frac{\partial J}{\partial w_0} = \frac{1}{n} \sum (\hat{y}_i - y_i)$$

$$\frac{\partial J}{\partial w_1} = \frac{1}{n} \sum (\hat{y}_i - y_i) x_i$$

2 Effect of Learning Rate

The learning rate r significantly affects convergence:

- **Small learning rate** ($r = 0.00001$): The algorithm converges slowly but steadily toward the optimal solution.
- **Moderate learning rate** ($r = 0.0005$): Faster convergence with stable updates.
- **Large learning rate** ($r = 0.001$): The algorithm may diverge, causing the loss to increase rapidly and parameters to become unstable.

In this experiment, a moderate learning rate ($r = 0.0008$) was chosen to ensure stable convergence.

3 Result Visualization

The final model is plotted together with the dataset points. The learned regression line approximates the relationship between x and y .

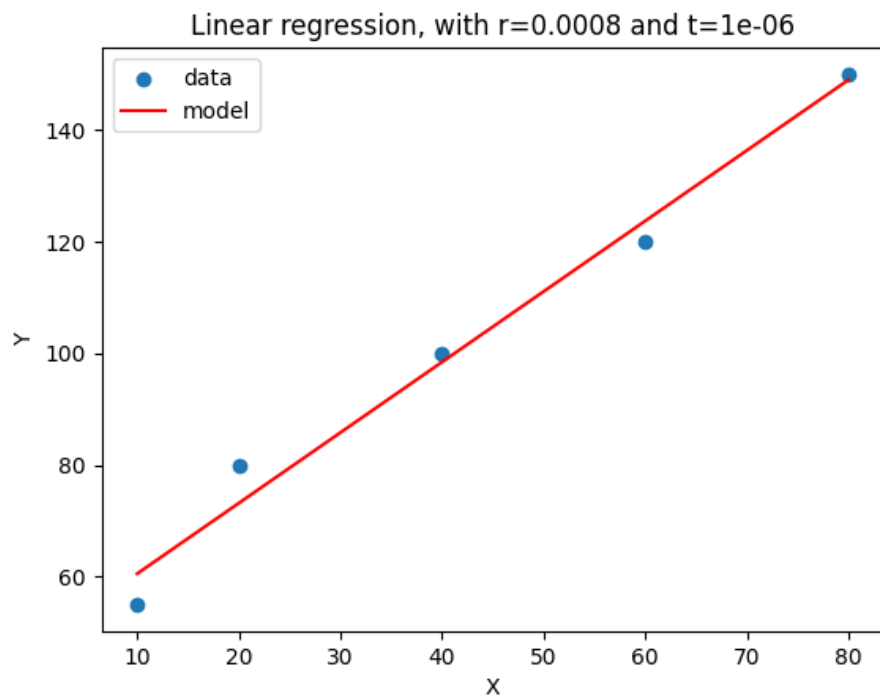


Figure 1: Linear regression plot